

1. A healthcare startup wants to build a model to predict the onset of diabetes. The dataset has 50 features, but the team wants to use only the most relevant ones. How can they apply feature selection effectively?

Feature & Model selection: selectKbest, classification

Step 1: Collect diabetes dataset which contains details like symptoms, medicines etc

Step 2: Preprocess the data handling null values & encode categorical data

Step 3: use feature selection method like selectKbest to extract important features or columns that can be used for ML model building

Step 4: Create a ML model using random forest or decision tree to predict the disease

Step 5: Evaluate the model using classification matrix, ROC-AUC, accuracy & precision

Step 6: Deploy the model using pickle to see results

2. You're creating a new project for managing student data. The project name is student_portal. What is the step-by-step command to start it? write logic for this django project

Command: `django-admin startproject student_portal`

Step 1: Open terminal

Step 2: Navigate to the required directory using `cd`

Step 3: Run the following command: `django-admin startproject student_portal`

Step 4: This creates a folder for student_portal

Step 5: Create sub folders & html files to execute the application

3. A company is analyzing customer purchase patterns with 200+ behavioral features. How can they reduce dimensionality without losing predictive power?

Feature & Model selection: LDA, classification

Step 1: Collect customer dataset which contains details like product, prices etc.

Step 2: Preprocess the data handling null values & encode categorical data

Step 3: use dimensionality reduction method like LDA to extract important features or columns that can be used for ML model building

Step 4: Create a ML model using random forest or decision tree to predict the purchase

Step 5: Evaluate the model using classification matrix, ROC-AUC, accuracy & precision

Step 6: Deploy the model using pickle to see results

4. A digital library wants to recommend books to readers based on what similar readers liked. How should they design this system?

ML Model: Recommendation system - Collaborative filtering

Step 1: Create user based interaction matrix from user ratings

Step 2: Calculate similarity between user rating (user based collaborative filtering) using cosine similarity

Step 3: Recommend books that similar users liked which target users haven't read

Step 4: Evaluate using RMSE or Recall

5. A bank wants to assess the risk level of credit applicants using only the most important financial indicators. How can they reduce the number of features?

Feature & Model Selection - RFE, Classification

Step 1: Collect applicants' dataset which contains details like credit score, employment etc.

Step 2: Preprocess the data handling null values & encode categorical data

Step 3: use feature selection method like RFE to extract important features or columns that can be used for ML model building

Step 4: Create a ML model using random forest or decision tree or Naive bayes to predict the risk level

Step 5: Evaluate the model using classification matrix, ROC-AUC, accuracy & precision

Step 6: Deploy the model using pickle to see results

6. A news app wants to recommend articles based on both article similarity and user reading history. How can they implement a hybrid system?

ML Model: Recommendation system - collaborative filtering

Step 1: Collect user behaviour information and the articles data

Step 2: Build content based recommender on article content

Step 3: create collaborative filtering of user-article interaction matrix

Step 4: Evaluate using CTR & time spent on site

Step 5: Retrain for every articles added

7: You're building a spam detection model and have thousands of text features from emails. How do you identify the most useful ones?

Feature & Model Selection - PCA, Classification

Step 1: Collect mail dataset which contains details like from, subject, spam/not spam etc.

Step 2: Preprocess the data handling null values & encode categorical data

Step 3: use feature selection method like PCA to extract important features or columns that can be used for ML model building

Step 4: Create a ML model using random forest or decision tree or Naive bayes to predict the spam emails

Step 5: Evaluate the model using classification matrix, ROC-AUC, accuracy & precision

Step 6: Deploy the model using pickle to see results

8. An ed-tech platform wants to recommend courses based on what similar learners have enrolled in. What steps would you take?

Model type: Recommendation system - collaborative filtering

Step 1: Create user based interaction matrix from courses selection

Step 2: Calculate similarity between user rating (user based collaborative filtering) using cosine similarity

Step 3: Recommend courses that similar users liked which target users haven't read

Step 4: Evaluate using RMSE or Recall

Step 5: Retrain the model for every course added

9. You're developing a car price prediction tool. With 100+ features (e.g., brand, mileage, engine type), how do you reduce complexity?

Feature & Model Selection - RFE, Regression

Step 1: Collect mail dataset which contains details like from, subject, spam/not spam etc.

Step 2: Preprocess the data handling null values & encode categorical data

Step 3: use feature selection method like RFE to extract important features or columns that can be used for ML model building

Step 4: Create a ML model using linear regression or random forest or SVM to predict the car price

Step 5: Evaluate the model using RSqaure, SSR, SSE

Step 6: Deploy the model using pickle to see results

10. How do you recommend products to new users who haven't interacted with anything yet?

ML model - Recommendation system - popularity based

Step 1: Create user based interaction matrix from products

Step 2: Calculate similarity between users (user based collaborative filtering) using cosine similarity

Step 3: Recommend products that similar users liked which target users haven't read

Step 4: Evaluate using RMSE or Recall

Step 5: Retrain the model for every products added